

Deconstructing glottalization, ejectives, and breathy voice in Armenian dialects

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Abstract: Previous work has reported that several Armenian varieties show ejective or glottalized stops, a claim with significant implications for the reconstruction of the Proto-Indo-European stop system. This paper reviews the historical, dialectological, and phonetic evidence for ejective, glottalized, and breathy-voiced stops in Armenian, finding that these claims largely rest on unreliable impressionistic reports. To provide an empirical assessment, we present preliminary acoustic data from field recordings of three native speakers of the Janik subdialect of Hamshen Armenian. This variety has previously been reported to possess glottalized/ejective stops, and we additionally test whether its historical voiced series is realized with breathy phonation. We conducted an exploratory acoustic analysis by extracting fundamental frequency (F0) trajectories and Cepstral Peak Prominence (CPP) values at the vowel onset following target plosives. Our initial findings yield no clear evidence of full glottal closure or severe aperiodicity associated with glottalization or ejectivity, nor of the laryngeal slackness associated with breathy voice. Instead, the target tokens predominantly exhibit high-falling F0 trajectories and CPP values characteristic of modal voice. While further large-scale articulatory and statistical work is needed, these preliminary results suggest that where non-modal phonation or ejectivity does occur in Armenian, it is likely sporadic or paralinguistic, offering little support for the existence of a stable ejective, glottalized, or breathy-voiced series.

Keywords: Armenian, breathy voice, consonants, dialectology, ejectives, Glottalic Theory, glottalization, phonetics

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К деконструкции глоттализации, эйективности и придыхательной фонации в армянских диалектах

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Аннотация: В предыдущих исследованиях отмечалось, что в ряде армянских идиомов представлены эйективные или глоттализированные смычные согласные. Это утверждение имеет значительные последствия для реконструкции праиндоевропейской системы смычных согласных. В статье рассматриваются исторические, диалектологические и фонетические свидетельства в пользу существования эйективных, глоттализированных и звонких придыхательных смычных в армянском языке. Показано, что эти утверждения по большей части опираются на ненадежные импрессионистические наблюдения. В статье анализируются акустические данные, полученные из записей трех носителей джаникского говора амшенского диалекта армянского языка. Согласно существующим исследованиям, в данном идиоме представлены глоттализированные/эйективные смычные, а историческая серия звонких придыхательных сохраняет придыхательную фонацию. Предварительный анализ траектории частоты основного тона (F0) и выраженности кепстрального пика (CPR) в начале гласного, следующего за взрывным согласным, не выявил признаков полного закрытия голосовой щели или выраженной аперiodичности, связанных с глоттализацией и эйективностью, равно как и гортанной расслабленности, характерной для придыхательной фонации. Напротив, данные преимущественно демонстрируют нисходящие траектории F0 и значения CPR, типичные для модальной фонации. Несмотря на необходимость дальнейших более масштабных артикуляционных и статистических исследований, полученные результаты позволяют предположить, что проявления немодальной фонации или эйективности в армянском языке, вероятнее всего, носят спорадический или паралингвистический характер, что не дает достаточных оснований для постулирования серии эйективных, глоттализированных или звонких придыхательных согласных.

Ключевые слова: армянский язык, диалектология, глоттальная теория, глоттализация, консонантизм, придыхательность, фонетика, эйективы

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1. Introduction

One of the central debates in Indo-European historical phonology concerns the structure of the Proto-Indo-European (PIE) stop system—specifically, whether the series traditionally represented as *D were plain voiced [b], [d], [g], [gʲ], [gʷ] [Mayrhofer 1986] or whether they were in fact voiceless and glottalized [pʰ], [tʰ], [kʰ], [kʰʲ], [kʰʷ] [Gamkrelidze, Ivanov 1973; Hopper

1973; Kortlandt 1985]. Armenian is of particular relevance to this debate, as many scholars have claimed that one or more modern varieties of the language exhibit a glottalized series [Hübschmann 1876; Kortlandt 1978: 13; 1985; 1995; 1998; 2003: 20–25, 126–128; 2010: 57–61; Chirikba 2008b: 43].

In this paper, we examine the nature, chronology, and geographic distribution of the putative glottalized series in Armenian—what these stops are, when and where they are attested, and how they have been interpreted. We then consider why so many researchers have taken Armenian evidence to support the existence of glottalized or ejective stops, and identify problems with many of those arguments. Our primary empirical focus is the small body of existing instrumental phonetic work on Armenian stops, which we reassess in light of newer articulatory and acoustic findings. To the limited extent that glottalization may exist in modern Armenian—in dialects from the Caucasus including that of Tiflis—we argue that it likely represents contact with neighboring Kartvelian and Northeast Caucasian languages, not inheritance from Proto-Armenian or PIE.

Building on these data, we present novel evidence suggesting that the relevant Armenian stops are best analyzed not as ejective or glottalized, but as plain or murmured voiced segments.

It is worth noting at the outset the difference between the easily conflated terms “ejective”, “glottal”, “glottalic”, and “glottalized”. While “glottal” can refer to a broad range of phenomena related to the glottis (e.g. the glottal stop [ʔ] or glottal vibration), consonants characterized as glottalic are produced specifically with a glottalic airstream mechanism, cf. [Gordeeva, Scobbie 2013]. The latter involves either the upward movement of the larynx combined with egressive airflow, used in ejective sounds, or its downward movement with ingressive airflow, producing implosives [Laver 1994: 171; Ladefoged, Johnson 2014: 314–315]. Furthermore, ejectives must not be confused with glottalized (or, more specifically, pre- and post-glottalized) sounds. This distinction mirrors the difference between any primary articulation and a secondary articulation (cf. palatal vs. palatalized, velar vs. velarized, etc.). However, the term “glottalization” (and similarly “laryngealization”) has been used “in diverse and ambiguous ways” in the linguistic literature [Esling et al. 2005: 390], a problem evident in its definition as “a very broad term applied loosely to a wide range of phenomena involving some kind of activity in the glottis” [Trask 1996: 160].

In our own instrumental study (see Section 3.4.1), we maintain the definition of glottalization strictly as a secondary articulation involving a glottal stop [Esling et al. 2005: 390], whereas the term “ejective” is used only for sounds produced with the larynx as the initiator of airflow. However, the original terminology will be retained when discussing historical literature, as the lack of articulatory detail or phonological interpretation rarely makes it possible to identify whether an author is referring to a primary or secondary articulation.

How does Armenian contribute to the broader Indo-European debate? In the standard framework established by Heinrich Hübschmann in the 1870s, Classical Armenian (the earliest attested stage of the language) exhibits systematic and seemingly straightforward correspondences among the three Proto-Indo-European stop series [Hübschmann 1876; Meillet 1936: 26–29]: Series I stops are realized as aspirated voiceless segments; Series II stops are plain voiceless; and Series III stops are plain voiced.

The modern dialects reveal a more complicated picture, however, with potential consequences for our reconstruction of the PIE stop system. Table 1 depicts, with representative varieties that instantiate them, the range of correspondences between the three PIE series and their outcomes in the modern dialects, which Armenian dialectologists typically categorize in seven groups (see [Pisowicz 1976] and [Vaux 1998: 9] for discussion). In this categorization, Classical Armenian belongs to Group 6, along with modern varieties such as Agulis and Standard Eastern Armenian.

In light of Table 1, let us quickly consider the three series in turn. Series I is consistently realized as voiceless aspirated in word-initial position across all Armenian varieties, and there-

Table 1

Armenian outcomes of the PIE stop series in word-initial position

	Series I	Series II	Series III
PIE	*T ↓	*D ↓	*Dh ↓
1. Sivas	Th	D	Dh
2. Yerevan	Th	T	Dh
3. Istanbul	Th	D	D
4. Sasun	Th	D	T
5. Malatia	Th	D	Th
6. Classical Armenian, Agulis	Th	T	D
7. Van	Th	T	T

fore will not be considered here. Series III, on the other hand, is interesting from an Indo-European perspective because it surfaces as voiced aspirated in a number of dialects such as New Julfa, Mush, Erzerum, and Akn, as first observed by Adjarian [Ačařean 1899; Adjarian 1899], Sievers [1901], and Maxudianz [1911] (cf. feature no. 2 in Jahukyan [1972] for an extensive list of such dialects). Classic Indo-Europeanist treatments of the Armenian stop system (e.g. [Hübschmann 1876; Meillet 1936]) postulated that this series was voiced and aspirated in PIE, and lost its aspiration in Classical Armenian (cf. Table 1). On the basis of the aforementioned dialects with voiced aspirate outcomes of Series III, though, scholars such as Pedersen [1904], Łaribyan [1953], Vogt [1958], Benveniste [1959], and Garrett [1991] have suggested that these dialects (and hence Proto-Armenian) may have preserved the original laryngeal settings of the corresponding Indo-European series. We evaluate the competing analyses of Series III elsewhere [Sayeed, Vaux 2026].

The final consonant series to consider, Series II, is the focus of the present study. This series has been reported to be voiceless glottalized in certain varieties of Armenian—an observation of particular significance to proponents of the Glottalic Theory, who view such reflexes as direct continuations of the original Indo-European state of affairs. In Table 2 we schematize the Glottalic Theory reconstruction of the PIE stop system, and the (putative) outcomes of the three PIE stop series in certain modern Armenian dialects that proponents of the Glottalic Theory adduce in support of it.

Table 2

Alternative presentation of Armenian stop series that is taken by its proponents to support the IE Glottalic Theory

	Series I	Series II	Series III
PIE (Glottalic Theory)	*T ↓	*T' ↓	*D(h) ↓
Janik Hamshen	Th	D	T'
Yerevan	Th	T'	Dh
Tiflis	Th	T'	D

Our principal aim in the present study is to assess how the contemporary Armenian dialects align, in light of the descriptions provided by Sievers [1876/2015] and subsequent scholars, with the claims of the Glottalic Theory that Series II stops were ancestrally glottalized, rather than plain voiced as in the standard reconstruction in Table 1. If the characterization in Table 2 of Series II in the Yerevan dialect by Sievers [1901: 141] and others is phonetically and phonologically accurate, it may bear directly on the validity of the Glottalic Theory: if Yerevan Series II really is glottalized, this could be interpreted as the retention of an originally glottalized Series II. This potential relevance helps to explain why these dialects have attracted sustained attention—and a considerable body of commentary—from Kortlandt and other linguists working within the Glottalic framework.

2. The putative glottalized series

2.1. The Sievers account

Earlier nineteenth-century observers (see [Akhverdyan 1852; Lepsius 1863]) did not identify ejectives in Armenian, by contrast with the ejectives that Lepsius had previously discussed in Ossetic. The first explicit claim of somewhat similar to the glottalic airstream mechanism in Armenian comes from Sievers [1876/2015], who worked with Georg Abulian, a speaker of the Tiflis dialect.

As quoted by Hübschmann [1876: 55–56], Sievers describes the T series in Abulian's dialect as follows:

The *tenués* differ from the usual European types in that they are produced with an actual closure of the larynx. The larynx rises noticeably with each articulation—by approximately half to three-quarters of an inch—to create the ensuing burst. The resulting acoustic effect is markedly distinct from that of the unaspirated *tenués* found in European languages (Slavic, Romance, etc.), which are not accompanied by laryngeal closure. The burst itself, perceived as a kind of scratching sound, is often more clearly separated from the following vowel than in European realizations, occasionally producing what amounts to a perceptible pause between the consonant and the vowel.

In essence, Sievers was describing a consonant involving two closures: one glottal and one oral. In a later letter to his contemporary Hugo Schuchardt, Sievers [1896/2015] clarified that the Armenian T' series of Tiflis differed from the corresponding Georgian series spoken in the same city. In Georgian, he observed a noticeable temporal lag between the release of the two closures, whereas in Armenian the releases appeared to occur almost simultaneously.

Sievers' account is particularly noteworthy because it includes direct observation of laryngeal movement—he reportedly watched the larynx rise and fall externally during articulation. This constitutes the only articulatory evidence ever recorded for this series in any Armenian variety. However, Sievers does not appear to have employed any experimental phonetic techniques to investigate this consonant series. Beyond his visual observations of laryngeal motion, therefore, the empirical foundation of his claims remains limited.

The unresolved issues in Sievers' account leave open the question of what exactly constitutes the purported glottalized/ejective T' series in Armenian. Seyfarth and Garellek [2018], in their comprehensive phonetic review of this topic, note that the term “glottalization” in the Armenian context has been applied to two distinct phenomena. The first involves glottal constriction produced with a pulmonic airstream mechanism, as proposed by Pisowicz [1997]. The second interpretation identifies the feature as ejective articulation, employing a glottalic airstream mechanism [Allen 1950; Ladefoged, Maddieson 1996: 67; Pisowicz 1998; Baronian 2017].

The difficulty with both interpretations is the near-total absence of instrumental evidence. Even Allen's influential 1950 study of an Eastern Armenian speaker from Abadan, Iran, relied entirely on auditory impression rather than articulatory or acoustic measurement. While his work provided phonetic data on aspiration and other details of pronunciation, it did not furnish empirical confirmation of glottalization.

2.2. Chronology and distribution

2.2.1. Dialect distribution

Sievers' description of Abulian's Tiflis Armenian is the earliest explicit statement that any Armenian variety possessed glottalized or "larynx-raised" stops. In both his 1876 and 1901 accounts he assigns this articulation specifically to the Tiflis dialect, emphasizing that the glottal constriction produced a "very short and sharply repelled" release [Sievers 1901: 177]—phonetically comparable, in his view, to the Georgian ejectives, with which Tiflis Armenians were in daily contact.

Sievers' Tiflis data became the template for later writers who inferred, typically without phonetic evidence, that similar glottalized stops existed elsewhere in Armenian. The diffusion of this idea far outstripped the empirical base: most twentieth-century references to "Armenian ejectives" do not specify a dialect, and several authors (see, e.g. [Karst 1901: 30; Hirt 1905: 138; Prokosch 1917; Vogt 1954: 371; Jensen 1959: 17]) simply assert that Armenian has ejectives, without argument or citation. These unsupported repetitions created an illusion of wide distribution, which the primary data do not sustain. Figure 1 shows the locations associated with Armenian dialects that have been claimed to possess a glottalized series. The outlined area shows the area in which Armenian has been in contact with other languages of the Caucasus, while the arrows show historical movements of Armenian-speaking communities between the relevant locations.

Closest to Tiflis, several dialects on the Caucasian linguistic frontier have at various times been said to exhibit glottalized stops. Tomson's descriptions of the Akhaltskha and Tiflis dialects [Tomson 1887; 1890] refer to glottal or "firm" articulation (հաստաձայն *hastadzayn*), and [Mseriants 1897] echoes this claim for Tiflis and Erzurum. Taken together, these observations establish Tiflis as the only dialect for which nineteenth-century authors consistently reported glottalization. Beyond Tiflis, however, the picture becomes more complex, and several oft-repeated claims dissolve under closer scrutiny.

The northeastern Anatolian dialect of Artvin is sometimes listed among the Armenian varieties thought to have a glottalized series, due largely to a citation chain that begins with [Cereteli 1968] and is repeated in later secondary sources. Chirikba [2008b: 43], summarizing Cereteli, states that Artvin Armenian displayed tense or glottalized pronunciations, and elsewhere characterizes the Artvin stop system as "similar to that of Tiflis: a ternary system of stops and affricates—voiced, voiceless glottalized, and voiceless aspirated", adding that "glottalization and aspiration are not strong, unless under emphasis" [Chirikba 2008a: 63].

However, earlier and more comprehensive descriptions of the Artvin dialect do not report glottalization. Adjarian's major survey [Ačarean 1911], which provides detailed phonetic observations for numerous dialects, includes no such series for Artvin; Grigoryan [1957: 436] likewise describes only a two-way contrast of voiced vs. voiceless aspirated stops, with no mention of glottalization or ejectives. The balance of the primary evidence therefore weighs against the existence of a glottalized/ejective series in Artvin Armenian. The isolated references to "glottalization" in later summaries appear to derive either from impressionistic interpretations of emphatic articulation or from methodological transfer from Georgian phonetic categories.

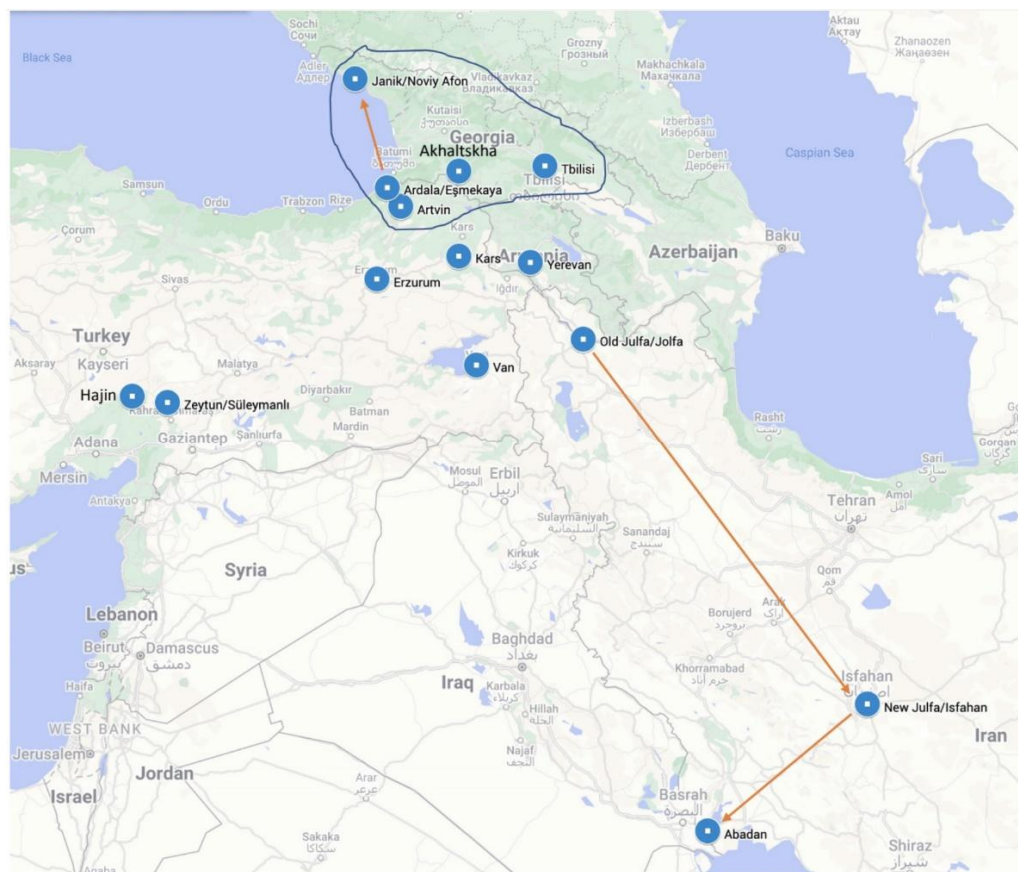


Fig. 1. Armenian dialects that have been claimed to have a glottalized/ejective series

Akhaltskha has likewise been claimed to possess glottalized stops based on Tomson's description of stops produced with the larynx closed [Tomson 1887]. No later phonetic sources confirm the presence of a glottalized series. In the absence of instrumental or independent descriptive corroboration, Tomson's remarks cannot be taken as evidence for ejectives in Akhaltskha; they are instead likely to reflect either fortis–lenis distinctions rendered in Caucasian-area descriptive terminology or an impressionistic attempt to capture a relatively tense voiceless articulation.

The Hamshen dialect, historically spoken in northeastern Anatolia, was transplanted in part to Abkhazia following the genocide, which began in 1915; the community has since been in sustained contact with languages of the Caucasus that have ejective series, including Mingrelian and Abkhaz. Outside of the dialects just mentioned, though, the remaining dialects claimed to possess a glottalized series are not from areas where contact with languages that feature ejectives or glottalized consonants would have been likely. For instance, in Cilicia, located in south-central Turkey, the dialect of Zeytun has been reported to exhibit a glottalic series [Khachaturian 1983: 60]. However, our research with speakers from these areas has not confirmed this phenomenon, though there may be various factors at play (see also [Açaryan 2003]). Other dialects claimed to contain the series include those of Van [Pisowicz 1997:229] and Erzurum [Sievers 1901:141] in southeastern and eastern Turkey, respectively, as well as Kars, a variety of the Erzurum dialect [Chirikba 2008:43], and Yerevan, the capital of present-day Armenia [Sievers 1901].

A significant number of authors claim that “Eastern Armenian”, or the literary standard, possesses a glottalized series [Weitenberg 2002: 147; Dum-Tragut 2009: 18; Maddieson 2013; Haig, Khan 2019: 8–9]. As with Western Armenian, none of these descriptions is based on instrumental data, and some directly contradict the findings of the two instrumental studies discussed below, namely [Schirru 2012; Seyfarth, Garellek 2018].

Specific claims about the Yerevan dialect vary widely. Pisowicz [1997] repeatedly asserts that Ararat/Yerevan T is glottalized, although his statements rely entirely on auditory impressions. Seyfarth and Garellek’s [2018] instrumental study of Yerevan Armenian, discussed in Section 3.3, found no evidence for ejective gestures. All robust instrumental studies converge on the conclusion that this variety lacks a phonologically glottalized series.

To reiterate, in each case reports rest on auditory impressions rather than on instrumental analysis, with no stable phonological pattern clearly emerging.

2.2.2. The illusion of wide distribution

A further group of authors simply presupposes the existence of glottalized/ejective consonants in Armenian without adducing primary evidence or even giving a source. In the Indo-European tradition, Karst [1901: 30], Hirt [1905: 138], and Prokosch [1917] treat Armenian as if it possessed a series of “ejectives” or “glottalized tenues,” but their discussions are purely classificatory: they do not specify which variety of Armenian they have in mind, they offer no phonetic description or example forms, and they do not cite any Armenian or Caucasian specialists. Later handbooks follow the same pattern. Vogt [1954: 371] and Jensen [1959: 17], for example, include an ejective/glottalized series in their phoneme charts for Armenian, yet again without indicating the dialectal basis or giving any argument beyond a brief label such as “glottalized” or “Caucasian-type.” These references are routinely recycled in the comparative and glottalic-theory literature as if they constituted independent confirmation of glottalization or ejectives in Armenian, but they in fact amount to bare assertions.

Areal and typological works are similar: [Haudricourt 1975: 267; Catford 1977: 70; Klimov 1986], and a range of more recent survey pieces, e.g. [Maddieson 2013; Bückins et al. 2018: 128; Aslanov 2018: 81; Stilo 2006], and [Donabedian 2018: 11], following [Haig 2001; 2014] and [Haig, Khan 2019], typically list Armenian alongside Georgian, Udi, etc., as having a ternary opposition of voiceless, glottalized, and voiced stops and affricates. In these texts, Armenian’s “glottalized”/“ejective” series often appears as a single line in a table or a brief remark in an areal overview, with no differentiation between Eastern vs. Western Armenian or between specific dialects such as Tiflis, Erzurum, or Yerevan. When examples are given at all, they are drawn from generic citation forms (e.g. /p’/, /t’/, /k’/) rather than from documented field data, and the only implicit justification seems to be the perceived structural parallelism with Georgian, or with Kartvelian more generally. In practice, these authors appear to be repeating an areal stereotype—that “Caucasian” languages have ejectives and Armenian behaves like its neighbors—rather than reporting independent observations on Armenian itself.

3. Arguments for Series II being glottalized

3.1. Non-phonetic evidence

We now examine the linguistic arguments that have been advanced for analyzing Armenian Series II stops as glottalized, beginning with three strands of non-phonetic evidence.

3.1.1.1. Treatment of Russian loans

Pisowicz [1976] argued that Russian loanwords offer support for analyzing Eastern Armenian Series II stops as glottalized. His reasoning is as follows. If Eastern Armenian /p/, /t/, /k/ were truly plain voiceless unaspirated—as in Russian—then Russian /p/, /t/, /k/ should be borrowed straightforwardly as Eastern Armenian /p/, /t/, /k/. Yet in some loans, Pisowicz notes that Russian unaspirated /p/, /t/ appear as Armenian aspirated /p^h/, /t^h/:

- (1) Russian loans adapted with aspirate stops in Eastern Armenian [Pisowicz 1976: 18]
печь /pʲetʃ/ ‘oven’ → p^hetʃ^h
пуд /pud/ ‘pood’ → p^hut^h
монета /moˈnʲeta/ ‘coin’ → manet^h ‘rouble’

Pisowicz interprets such forms as evidence that the Armenian Series II could not have matched Russian /p/, /t/, /k/ phonetically; instead, he suggests the Armenian series must have been glottalized, making the aspirated Series I the closer perceptual match in borrowing.

However, the presence of glottalized stops in Series II would not preclude the borrowing of Russian /p/, /t/, /k/ into that series. Georgian borrows Russian /p/, /t/, /k/ as ejective, not because the Russian stops are ejective, but because the ejective series is the closest match in Georgian for a stop with short-lag VOT. If Eastern Armenian had a similar inventory to Georgian, with a glottalized Series II and an aspirated Series I, one could easily expect the same outcome for /p/, /t/, /k/ as in Georgian—making the failure of that outcome only weak evidence against glottalization in Series II.

Moreover, this claim is weakened by Khachatryan and Khachaturova’s [1979: 262] observation that these exceptional items most likely came into Armenian via Azerbaijani, not directly from Russian. Azerbaijani, like English, lacks a voiceless unaspirated series and instead uses its voiceless aspirated series when borrowing Russian voiceless stops. By contrast, direct loans from Russian to Armenian overwhelmingly map Russian /p/, /t/, /k/ onto Armenian /p/, /t/, /k/, with no aspiration¹: *потолок* /potoˈlok/ ‘ceiling, upper limit’ → *patalok*; *колбаса* /kolbaˈsa/ ‘sausage’ → *kalbas*; *тактика* /ˈtaktika/ ‘tactics’ → *taktika*.

Khachatryan and Khachaturova further note that Eastern Armenian stops and affricates are not normally glottalized. Adjarian [Ačarean 1911: 39, 52] notes that Armenian *p*, *t*, *k*, *c*, *č* are pronounced with a glottal closure in Tiflis, but in the Yerevan dialect these sounds lack such an articulation. Consequently, the increased articulatory force of the stops is not a property of Armenian as a whole, but only of those dialects—particularly the Tiflis dialect—that were in close contact with the Kartvelian languages. Gevorg Ĵahukyan also maintains that although the Armenian stops and affricates form a three-way system, their oppositions differ qualitatively from the corresponding contrasts in the Caucasian languages; see [Ĵahukyan 1967: 77–78]. Experimental data support this interpretation of the Armenian stops; see [Khachatryan, Ayrapetyan 1971; Khachatryan, Khachaturova 1979: 262].

In sum, the Russian-loan evidence that Pisowicz takes as indicative of a glottalized/ejective Series II in Eastern Armenian evaporates under closer scrutiny: the key forms are better explained as Azerbaijani-mediated borrowings, and instrumental studies demonstrate that Yerevan Armenian, which forms the base of Standard Eastern pronunciation, lacks the kind of ejective articulation or glottalization characteristic of Kartvelian or Northeast Caucasian systems.

¹ Voiceless affricates by contrast are normally borrowed as aspirated, perhaps due to the spread glottis of the sibilant release: *чемодан* /tʃemoˈdan/ ‘suitcase’ → tʃ^hamadan, *чай* /tʃaj/ ‘tea’ → tʃ^haj, *царь* /tsarj/ ‘tsar’ → ts^har, *цыган* /tsiˈgan/ ‘gypsy’ → ts^higan, etc. One often finds the same behavior with stop-sibilant sequences, e.g. *такси* /takˈsi/ ‘taxi’ → tak^hsi.

3.1.2. Dialects outside of the Caucasus with glottalized T'

A second strand of evidence invoked by Pisowicz [1997] and Kortlandt [1978] concerns claims that certain Armenian dialects spoken outside the Caucasus exhibit a glottalized series, as represented on the map in Figure 1. At face value, such reports would appear to weaken the view that ejective-like articulation in Armenian is strictly an areal feature limited to the Caucasus. However, closer examination reveals at least two issues with this argument.

First, the relevant descriptions were based on work with speakers said to originate from communities such as Hajin, Zeytun, and Van—western Armenian regions where no contact with Caucasian languages possessing ejectives is historically attested. Crucially, though, the speakers consulted in these studies were not born and raised in the original western Armenian locations, but rather were descendants of refugees who settled in and around Yerevan during the genocide of 1915–1924. Since that time the descendants of the original refugees have lived in areas where the ambient variety is something close to Standard Eastern Armenian, a variety that the same scholars claim to possess some degree of glottalization in its voiceless series. If their claims are correct, distinguishing inherited features of the refugees' source dialects from features acquired through long-term accommodation to the ambient Caucasian speech community becomes extremely challenging.

Compounding this problem is the well-documented difficulty of eliciting “heritage” or “grandparents' dialect” forms from speakers whose everyday linguistic practice has shifted toward the local standard. In the absence of targeted sociophonetic elicitation procedures, there is a significant risk that what is interpreted as an archaic glottalized series is in fact a reflex of the speakers' long-term residence in an area where glottalization may itself be a contact-induced feature—possibly spreading from Tiflis during the period in which the standard language center shifted from Tiflis to Yerevan.

Consequently, the claim that dialects outside the Caucasus possess an inherited glottalized/ejective T' lacks firm empirical grounding. The available evidence is at least equally compatible with a scenario in which local Caucasian phonetics have influenced originally refugee communities over several generations, thereby creating an illusion of glottalized consonants in dialects that, in their original homelands, did not possess such a feature.

3.1.3. Avoidance of exchange rule $D \leftrightarrow T$

Another non-phonetic argument by Kortlandt [1978] for reconstructing a glottalized/ejective T' series in Proto-Armenian is that it would eliminate the apparent exchange rule seen in the Group 4 dialects with the inventory T^h –D–T, where Classical Armenian D corresponds to Group 4 T, and Classical Armenian T corresponds to Group 4 D. Many linguists consider true “flip-flop” exchanges ($X \leftrightarrow Y$ in one step) to be impossible. According to Kortlandt, positing original glottalized/ejective T' avoids this problem: the original plain voiced series (D) first shifts into the T slot, and only afterwards does the glottalized/ejective T' move into the vacated D position. In that scenario, no two categories swap directly; rather, they change sequentially.

The logic of Kortlandt's argument presupposes (i) that the Armenian T-series was glottalized/ejective in the proto-language, and (ii) that the later appearance of glottalization or ejectives in many dialects reflects inheritance rather than the well-attested Caucasian areal effect. But both assumptions are problematic.

First, there is no unambiguous evidence for inherited glottalization or ejectives in Armenian, and most of the pre-instrumental claims reviewed in Section 2.2.2 give no data or references, or simply rest on the typological expectation that Armenian “ought” to look like Georgian. The instrumental studies [Allen 1950; Khachatryan, Ayrapetyan 1971; Schirru 2012; Seyfarth, Garellek 2018] find either no glottalization/ejectives or only sporadic, speaker-dependent creakiness, and in Yerevan Armenian the number of true ejectives is fewer than five tokens out of

225 [Schirru 2012]. In other dialects, including Tiflis and Janik, the glottalization or ejectives that have been reported is exactly of the kind expected in the northern Caucasus Sprachbund [Chirikba 2008b; Haig, Khan 2019], where Armenian, Kurdish, Neo-Aramaic, Laz, and Ossetic have all developed weak glottalization or ejectives through contact. In short, there is no phonetic or dialectological basis for assuming Proto-Armenian possessed an inherited glottalized series capable of feeding Kortlandt's scenario.

Second, the $D \leftrightarrow T$ exchange itself does not require a single-step flip-flop, and thus does not support the need for glottalized/ejective T' in the proto-language. As already noted by Weitenberg [2002], there are several regular, independently motivated, non-exchange pathways capable of deriving Group 4 $/t^h/-/d/-/t/$ from the Classical Armenian $/t^h/-/t/-/d/$ system. To use an example of our own, one can derive Group 4 without any flip-flop:

- (2) A diachronic derivation of Group 4 without flip-flop:
 Step 1: $/d/ \rightarrow /d^h/$ (breathy voicing), as in Groups 1–2 (e.g. Sivas and Yerevan);
 Step 2: $/t/ \rightarrow /d/$ (voicing), counterfeeding Step 1, as in Group 2 (e.g. Yerevan);
 Step 3: $/d^h/ \rightarrow /t/$ (loss of breathiness + devoicing in one step).

This final step requires devoicing and deaspiration to occur simultaneously in $/d^h/$, but exactly this kind of cluster-loss is well attested in Armenian: the closure phase of the breathy-voiced stops has been described for decades as partially or fully voiceless [Allen 1950; Khachaturian 1983], which makes the shift $/d^h/ \rightarrow /t/$ phonetically natural. Crucially, no part of this sequence requires a simultaneous $D \leftrightarrow T$ exchange, and none of it requires reconstructing a glottalized/ejective T' in Proto-Armenian.

Finally, even if one accepted the glottalic reconstruction for Proto-Indo-European, nothing in the Armenian dialect data suggests that any inherited glottalization or ejectives survived into Proto-Armenian. The dialects where glottalization is strongest (Tiflis, and perhaps Akhaltskha and Artvin) are precisely those with intense Caucasian contact; dialects in which we would expect inheritance (e.g. New Julfa, Ghap'an, Nakhijevan, Van, Western Armenian generally) either show no glottalization/ejectives or show the weak, variable, non-ejective "tensed" articulation associated with areal diffusion, not retention.

In sum, Kortlandt's argument that reconstructing glottalized/ejective T' avoids a $D \leftrightarrow T$ exchange is unnecessary on historical grounds and unsupported on phonetic grounds. The apparent problem disappears once the changes are decomposed into ordinary sequential developments, and the glottalization that Kortlandt takes as ancestral is best explained as a later, geographically localized Caucasian areal feature. Thus the Armenian evidence does not motivate the glottalic reconstruction—in fact it weakens the case for it—and offers no reason to posit an exchange-avoiding T' series in Proto-Armenian.

3.2. Pre-instrumental reports

In the phonetic domain, the primary evidence adduced for a glottalic series in Armenian has consisted almost entirely of impressionistic auditory reports. Sievers' observations, discussed above, belong to this category. Tomson, working in Georgia in the 1880s and 1890s, described the dialects of Tiflis and Akhaltskha as sounding somewhat like Georgian, a judgment based solely on ear rather than on instrumental measurement. Allen's 1950 description of a speaker of New Julfa Armenian from Abadan echoes this pattern, again relying on auditory impressions. By contrast, Adjarian, whose 1911 survey remains the most extensive treatment of Armenian dialects, confirmed glottalization only for Tiflis, consistent with our interpretation that glottalization in Armenian today is a result of areal contact.

In all these cases, the linguists were simply reporting their impressions of the speech of individual consultants; none of them provided instrumental confirmation. Sievers' visual obser-

vations of laryngeal movement, discussed above, fall short of providing controlled phonetic evidence. Adjarian carried out instrumental studies of several Armenian dialects while based in Paris in the 1890s, but none of these varieties (Noux and Mush [Adjarian 1899]) possessed a glottalized/ejective series, nor were claimed to.

3.3. Instrumental reports

The final—and most important—component of the present argument concerns the instrumental phonetic evidence. Of nearly a hundred publications that assert or assume the existence of a glottalized series in Armenian, only three investigate the question instrumentally, and of these only two studies emerge as genuinely informative.

The first such study is [Schirru 2012], based on data elicited from five speakers of the Yerevan dialect. Schirru recorded 225 tokens representing the dialect's three stop series, traditionally described as voiced (D), voiceless (T), and voiceless aspirated (Th). To determine whether the voiceless unaspirated series (T) showed glottalization, he examined three spectral-tilt measures—H2–H1, A1–H1, and A3–H1. In addition to these instrumental measures, he also listened for audible ejectivity. Out of 225 word-initial /t/-tokens, he judged four or fewer to sound ejective in the Georgian sense—that is, displaying a double release reflecting sequential oral and glottal bursts.

There are at least three problems for the reader hoping to take away from Schirru's report the conclusion that Yerevan has a glottalic series. First, Schirru's identification of ejectives is impressionistic, not instrumental, placing it in the same methodological category as Allen's earlier reports. Second, by Schirru's own report, the instrumental study that he **did** conduct (looking at spectral tilt) suggests that the differences among the three Yerevan series reflect distinct voice qualities, not systematic glottalization or ejectivity. In other words, Schirru's instrumental data provide no evidence for a robust or consistent glottalized/ejective T series in Yerevan Armenian. Finally, as Seyfarth and Garellek [2018] point out, spectral tilt values alone cannot diagnose glottal constriction in this dialect, because the D series is breathy voiced. Breathily phonation elevates spectral tilt, meaning that simply comparing tilt values across series cannot reveal whether the T series is modal, glottalized or ejective.

Their approach begins from a phonetic assumption grounded in cross-linguistic work on ejectives: "For glottalized plosives, the voice quality of adjacent vowels becomes creakier when the glottal-constriction gesture associated with the stop closure overlaps with the adjacent segments, making glottal constriction perceptible near an otherwise silent closure" [Ibid.: 427]. If the release of the glottal closure is not perfectly synchronized with the onset (or offset) of voicing for the following (or preceding) vowel, creaky voice appears in the adjacent vocalic portion—an effect that can be measured instrumentally.

To test for such cues, Seyfarth and Garellek recorded eight Yerevan speakers, each producing 111 frequency-balanced monosyllabic forms beginning with a voiceless stop, followed by a vowel and a single coda consonant. Their analysis considered three key measures: (i) H1–H2 spectral tilt, the difference between the amplitude of the first two harmonics in the spectrum, indexing voice quality; (ii) Cepstral Peak Prominence (CPP), a measure of harmonic-to-noise ratio that correlates with both aspiration noise and irregular vocal-fold vibration (creak); and (iii) F₀ onset, used to diagnose what they ultimately term "tense voice".

The authors plot CPP against H1–H2 and compare the three series directly to one another. If the voiceless T series involved systematic glottal constriction or ejectivity, it should exhibit lower H1–H2 values than both of the other series—reflecting increased constriction—together with a comparably low CPP, indexing irregular or creaky voicing in the adjacent vowel.

Their sample consisted of eight speakers (six female, two male). For all six female speakers, they found no acoustic evidence of glottal constriction: neither the spectral measures nor the

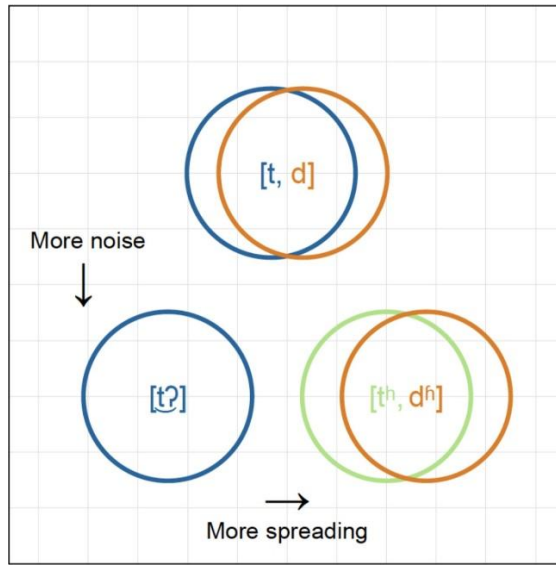


Fig. 2. CPP and H1–H2 values for the three Yerevan Armenian stop series [Seyfarth, Garellek 2018: 428]

waveforms themselves (manually inspected for irregular pitch periods) showed signs of glottalized onsets for the voiceless stops. For the two male speakers, however, the CPP and H1–H2 values deviated slightly. These patterns were consistent with tense voice, not ejectives. Tense voice, in their usage, is a subtype of creaky phonation involving increased vocal-fold constriction but accompanied by a higher F_0 , resembling that of the voiceless aspirated series rather than that of true ejectives.

The authors conclude that “although voice quality is difficult to perceive or to measure directly during an oral stop closure—and impossible during a voiceless one—future instrumental work might explore the timing and degree of glottal constriction via articulatory glottography during the closed interval” [Ibid.: 443]. This remark underscores the central limitation of all existing studies: direct articulatory evidence—especially for the glottis during closure—is still lacking. Further progress will require imaging techniques (e.g. electroglottography or high-speed laryngography) capable of capturing laryngeal gestures that are acoustically masked in voiceless stops.

The final phonetic study to consider is [Toparlak, Dolatian 2023], which again examines Yerevan Armenian. Unlike earlier works, this study incorporates electroglottography (EGG)—the closest we currently come to observing laryngeal behavior directly during stop production. Using the EVA 2 system, which employs a facial mask allowing simultaneous measurement of oral airflow, intraoral pressure, and laryngeal contact, they recorded three speakers of the Yerevan dialect and obtained 397 tokens across the three stop series (T, D, Th) in word-initial, medial, and final position. Only about one-third of these were T-series stops, and only one-third of those occurred in initial position.

Toparlak and Dolatian report that some speakers produced final /t/ as ejective, a finding entirely plausible typologically. Final-position glottal reinforcement is cross-linguistically widespread and is often independent of the presence or absence of a glottalized/ejective series in other positions, as they correctly note. It is therefore not relevant to our investigation of whether Armenian varieties possess or possessed a glottalized/ejective series.

A further problem is that although their study sampled stops in initial, medial, and final positions, Toparlak and Dolatian do not present intraoral-pressure or EGG data for initial stops,

which is precisely the environment crucial to evaluating claims of an initial glottalized T series. For that reason, this study, despite its methodological advantages, ultimately provides no direct evidence relevant to the question at hand.

Unfortunately, the Tiflis dialect, for which glottalization has been most consistently claimed, appears to be moribund or extinct, making direct instrumental investigation impossible. As stated above, the areal features of Caucasian languages suggest that glottalization or ejectives in Tiflis would not provide clear evidence of glottalization or ejectivity in Proto-Armenian.

3.4. Phonetic analysis of “ejectives” in the Janik Hamshen Armenian of Abkhazia

A more promising alternative to studying the now-extinct Tiflis variety is to examine non-standard dialects whose phonetic histories are less entangled with the sociopolitical shifts of the Armenian highlands. One such case is the Janik subdialect of Hamshen, now spoken by Armenian communities in Abkhazia. As noted earlier, Chirikba, himself a speaker of Abkhaz, states that Janik (a Hamshen dialect originally from northeast Anatolia, transplanted to Abkhazia more than a hundred years ago) “has a ternary system of stops and affricates: voiced, voiceless aspirated and voiceless unaspirated. The latter sounds are tense, and when put under emphasis (e.g., when a speaker is asked to repeat himself), they are pronounced with what sounds as weak glottalization, though it is not quite clear whether glottal (ejective) coarticulation is involved, e.g. *č.ehez* [tʰehez] ‘bride’s dowry’, *t.un* [tʰũn] ‘thou’, *p.er-im-gu* [pʰe'rimgu] ‘I (shall) bring hither’, etc.” (transcriptions original) [Chirikba 2008a: 58–59].

To begin assessing this claim, we asked three native speakers of the Janik variety—Avik (Novyj Afon), Gohar (Gantiadi)², and Amalia (Sukhumi)—to record these three items, repeating each of them three times with pauses in between. Representative spectrograms are given in Figure 3. The vertical axis represents frequency in Hertz.

3.4.1. General profile of the recordings

Across nearly all tokens from all three speakers, the acoustic evidence is inconsistent with full glottal closure during the stop or affricate. In each speaker’s productions, the release of the consonant is either followed by a vowel or some aspiration. The aspiration implies that the glottis is at least partially open at release, rather than tightly adducted as in ejective stops. The visual absence of severe glottalization (though see the vowel onset of [pe'rimgu] in Figure 3) is quantitatively supported by Cepstral Peak Prominence (CPP) metrics extracted across all three speakers; the target tokens yielded mean CPP values ranging from 11.36 to 12.27, indicating predominantly periodic, modal vowel phonation rather than the severe aperiodicity associated with ejectives. These properties do not align with the abrupt onset and high periodicity expected from an ejective or strongly glottalized articulation.

None of the tokens shows a silent or near-silent interval between the oral release and the onset of voicing, nor do they display the high-amplitude, sharply defined glottal bursts that accompany full glottal closure. No token exhibits the characteristic F0 spike at vowel onset typical of Caucasian ejectives. In all three speakers, therefore, the stops pattern more like lax segments, analogous to the phonatory settings attested in lax/tense contrasts in the world’s languages.

² Gohar states that her paternal grandfather was from Janik, her paternal grandmother from Ordu, and her mother’s side from Trabzon. Although these field recordings were not collected under controlled laboratory conditions, they nonetheless permit a first assessment of whether any of the productions exhibit acoustic correlates compatible with glottal closure, ejectivity, or systematic glottal constriction.

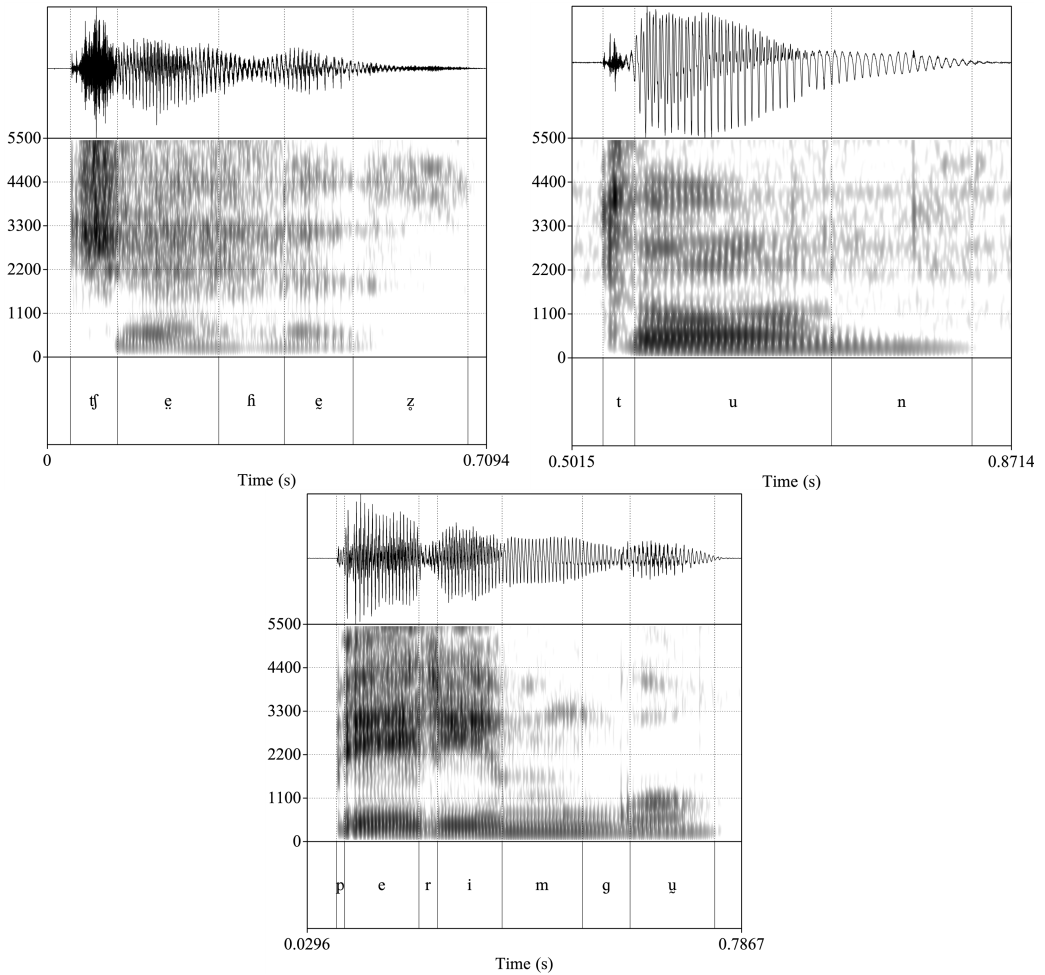


Fig. 3. The waveforms and spectrograms of [tʃe'fiɛz] 'bride's dowry', [tun] 'thou', [pe'rimqu] 'I (shall) bring hither' as produced by Gohar

3.4.2. Speaker-specific patterns

3.4.2.1. Avik and Amalia

Recordings from two of the speakers show no clear evidence of glottal closure. Both speakers show weak bursts, short VOTs, and breathy or modal voicing at the onset of the vowel. One of Avik's tokens shows a momentary weakening of the glottal pulses, which may better reflect a transient flapping of the vocal folds rather than any glottal closure. In both cases, the data are more consistent with lax voiceless articulations with brief aspiration, rather than ejectives.

3.4.2.2. Gohar

Gohar's recordings are more complex. Most of her tokens begin with modal or slightly lax voicing, but one—the form [pʰe'rimqu]—displays an onset with irregular, high-energy glottal

pulses. This creaky onset may reflect increased glottal tension or partial constriction, though Gohar's speech in general is characterized by the paralinguistic creak, perhaps connectable to her being bilingual in California English. Two small peaks in the waveform immediately following the burst could represent resonances in a nearly closed glottis, though this interpretation is speculative. One affricate is unusually long (about 70 ms), atypical for a glottalic affricate, and its low-frequency burst could reflect either lip rounding before [u] or partial constriction. Even so, the token lacks any silent closure interval, has no delayed onset of modal phonation, and does not exhibit the compression phase characteristic of ejectives. At most, these patterns indicate tense or creaky phonation, not a glottalic airstream mechanism. An aspirated ejective—something like [tʰ]—cannot be categorically ruled out acoustically, but nothing in the data demands such an analysis. This interpretation aligns with the finding from [Seyfarth, Garellek 2018] that certain Yerevan speakers produce T-series stops with a tense or creaky phonatory setting that does not involve a glottalic airstream mechanism.

Interestingly, Gohar produced a few likely word-initial ejectives in three other tokens, recorded for a different study, namely, *tunk* 'thou', *tus* 'out(side)', and *kov* 'thief'. The production in question is characterized by a period of silence following the burst, see Figure 4.

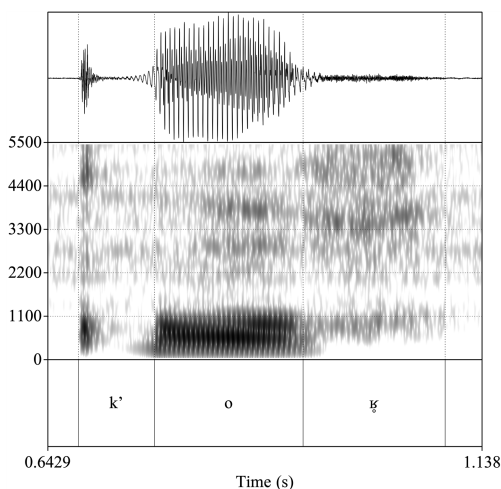


Fig. 4. The waveform and spectrogram of [k'og] 'thief' as produced by Gohar

However, since such articulations were observed sporadically (repetitions 1 and 3 for *tunk*, repetition 2 for *tus*, and all repetitions for *kov*), this example cannot be used to postulate a phonemic series of ejectives.

3.4.3. Interpreting phonetic evidence

In interpreting previous reports of glottalization, one must remain aware of listener bias. Researchers accustomed to Georgian-type ejectives may under-identify weaker or atypical forms of glottalization, whereas scholars sympathetic to the Glottalic Theory may be predisposed to interpret faint laryngeal gestures as glottalization. This consideration is especially pertinent in evaluating Chirikba's perceptual classification: as a speaker of Abkhaz and a scholar trained in the Leiden tradition, his perceptual templates and theoretical expectations may differ from those of phoneticians approaching the data from a cross-linguistic standpoint. Overall, the Janik data examined here show no acoustic evidence for full glottal closure, and only one token shows phonatory properties consistent with tense or creaky phonation. The recordings pre-

dominantly exhibit lax, modal, or breathy voicing, with weak bursts and short affrication—precisely the patterns found elsewhere in Armenian dialects where impressionistic claims of glottalization or ejectives have dissolved under instrumental scrutiny. While further articulatory work (ideally electroglottography or high-speed laryngography) will be necessary to reach a definitive conclusion, the present evidence suggests that Janik Armenian does not possess a system of glottalized/ejective stops. Its “tense” series appears to reflect phonatory tenseness rather than a glottalic airstream mechanism, and thus aligns with broader tendencies in Armenian for tenseness or breathiness to be misheard—or misinterpreted—as glottalization or ejectivity.

3.5. Phonetic analysis of “breathiness” in the Janik Hamshen Armenian of Abkhazia

In addition to the “tense” voiceless series, we must account for the phonetic realization of the historical voiced stops. In dialects such as Yerevan Armenian, utterance-initial voiced stops frequently lack phonetic prevoicing during the closure phase. To maintain the phonemic contrast, speakers rely on laryngeal cues at the onset of the following vowel—specifically, a slack vocal fold posture that produces breathy phonation and a distinctly depressed fundamental frequency contour [Seyfarth, Garellek 2018].

To determine if the Janik subdialect utilizes a similar micro-prosodic strategy, we analyzed the F0 trajectories following target stops historically categorized as voiced (cf. forms like /b^hon/ ‘thing, work’ or /g^hel/ ‘wolf’ that Adjarian [Ačaryan 1947] provides for Mala, whereas in Janik he reports plain voiceless initial stops). Because our recordings lack strict minimal pairs—introducing variability from intrinsic vowel pitch—absolute F0 values are less informative than the shape of the pitch contour. To account for different vowel durations, we time-normalized and extracted the mean F0 trajectories over the first 50% of the vowel across all three speakers.

3.5.1. Acoustic absence of laryngeal slackness

If the Janik stops maintained contrast via laryngeal slackness or breathy phonation, we would expect to see a rising F0 trajectory (a low “hook”) at the vowel onset as the slack vocal folds gradually tense to meet the intrinsic pitch of the vowel. The acoustic data, however, overwhelmingly contradicts this hypothesis.

As shown in Figure 5, the F0 trajectories universally exhibit a “stiff voice” contour. Across all target tokens for all speakers, the pitch begins high at the vowel onset (means ranging from roughly 158 Hz to 196 Hz) and undergoes a steady, gradual declination. A high-falling pitch contour at vowel onset physically precludes a slack vocal fold posture; it requires high longitudinal tension of the vocal folds, a physiological gesture typically associated with plain voiceless or fortis aspirated stops.

3.5.2. Interpreting phonetic evidence

The complete absence of pitch depression at the vowel onset indicates that these speakers do not employ breathy phonation or laryngeal slackness to mark these stops. In the absence of prevoicing, these segments pattern acoustically as plain voiceless or voiceless aspirated stops, showing no trace of the laryngeal artifacts found in the Yerevan variety.

Mean F0 Trajectories at Vowel Onset (Target: Breathiness / Slackness)

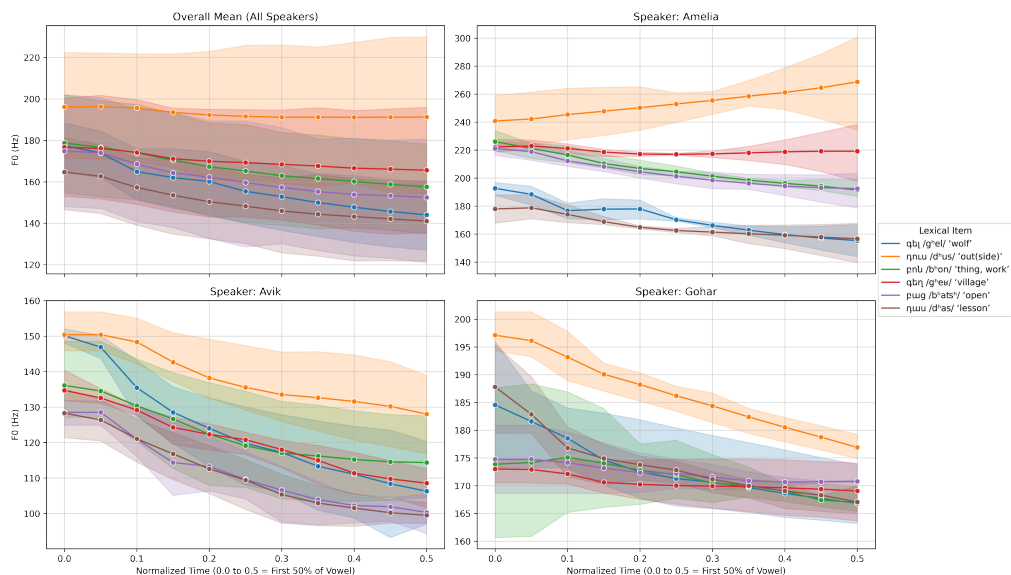


Fig. 5. Mean fundamental frequency (F0) trajectories across the first 50% of the vowel duration following target plosives, separated by speaker. Lines represent the mean across all repetitions per token; shaded areas indicate 95% confidence intervals. Pitch contours were time-normalized to 11 equally spaced intervals to account for variations in absolute duration and micro-aspiration at onset

4. Conclusion

The instrumental evidence gathered to date shows no clear support for the existence of a systematic ejective or glottalized T series in today's Armenian dialects. Where glottalization-like phonation does exist, we argue that it either represents an effect of contact with neighboring languages of the Caucasus, not from inheritance, or is sporadic and/or paralinguistic.

As for the historical breathy voiced series, we have found no indication of its presence in the speech of our Hamshen informants. The series in question seems to lack any prevoicing as well as any non-modal voice quality on the following vowel.

Several points follow from these results. First, any decisive progress will ideally require articulatory evidence. Direct laryngoscopic imaging would, in principle, be ideal, but in practice it is unrealistic: dialect speakers are often willing to make phone recordings, but not to undergo invasive clinical procedures. Electroglottography (EGG) is far more feasible, non-intrusive, and promising, though it still requires in-person recording and provides only indirect inference about glottal configuration rather than full visualization.

Second, future work should not be confined to Yerevan, whose stop system is unrepresentative of many dialects historically claimed to have glottalized/ejective series. We should aim to investigate non-standard dialects, especially those where contact histories are less complex than the deeply transformed linguistic ecology of Armenia and its diasporas. Such work must, where possible, control for diasporic influence, since many Armenian communities today are surrounded by speakers of either strongly glottalic or entirely non-glottalic languages—conditions that may distort inherited patterns.

Third, careful control of age is essential. Given the rapid erosion of traditional dialects in the contemporary period, younger speakers are far less likely to preserve older phonetic systems.

Control of gender is also necessary, as the apparent gender skew in Seyfarth and Garellek's findings illustrates; without systematic sampling, sociophonetic interpretation is problematic. Finally, degree of education must be accounted for, given the strong influence of the prescriptive Eastern Armenian standard—whose orthography and normative phonology explicitly lack glottalization/ejectivity and may shape speakers' productions.

In short, although the historical and impressionistic record is vast, the instrumental evidence to date provides no compelling support for the existence of an inherited glottalized/ejective series in Armenian. A meaningful resolution will require renewed, carefully controlled, and genuinely articulatory research across a broader range of dialects—particularly those in which glottalization/ejectivity has been claimed but never documented with modern phonetic methods.

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